

Data Science

Lecture 02: Probability Theory

Prof. Dr. Michael Granitzer

Converted from LNKB-tex part-en-ds-chp2-probability-and-statistics.tex. All twenty content units of the original chapter are included below, in the original order, as shared units from content/units/math/stochastic/ — the same units feed the probabilistic machine learning course.

Module 0

Orientation

Guiding Question: *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory

Given a model of chance,
what follows?

Kolmogorov axioms,
distributions, limit theorems

Mathematical statistics

Given data, which model of
chance is plausible?

Estimation, hypothesis testing

Data science

Given data and a decision,
what should we do?

Prediction, learning,
evaluation

The arrow runs left to right: this module builds the vocabulary the rest of the course spends its time using.

Orientation slide. The three columns are not three fields to memorise but three questions — deduction from a model, inference of a model, decision with a model. Say explicitly that we start on the left because everything on the right is stated in its language.

What We Need From Probability

In this module

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models

Set expectations: this is the one heavily formal module, and every item on the left has a named consumer on the right. If students ask "why do we need this", point at the right-hand column rather than defending rigour in the abstract.



Example: Two Fruit Boxes

Red box

2 apples, 6 oranges

Blue box

3 apples, 9 oranges

We pick a box, then a fruit from it.

- If we pick the **red box** **40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

Speaker notes

Deliberate overview slide (P4): the two questions are the whole chapter in miniature. Do not formalise anything yet — the point is that both questions are answerable by counting, and that the second one feels harder than the first.



The Counts, in One Grid

Twenty picks, recorded as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	2	3	5
$F = o$ (orange)	6	9	15
column total	8	12	20

Joint

a single cell over the total

$$P(B = r, F = a) = \frac{2}{20}$$

Marginal

a row or column total over the total

$$P(F = a) = \frac{5}{20}$$

Conditional

a cell over its own row or column

$$P(F = a \mid B = r) = \frac{2}{8}$$

Everything on this slide is division of counts. Have students compute one of each kind before naming them. The names will be redefined formally later here they are labels for three ways of dividing the same table.

The Two Rules

Sum rule

$$P(X) = \sum_Y P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y | X)P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

This is the Bishop framing: two rules, everything else derived. Students who carry only these two out of the chapter are in good shape — the formal apparatus that follows exists to justify them and to extend them beyond finite tables.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

Speaker notes

The example is deliberately degenerate: identical ratios in both boxes make the posterior equal the prior. That is a stronger lesson than a numerical surprise — evidence updates belief only to the extent that hypotheses disagree. Change the red box to 1 apple / 7 oranges and rerun if the group wants to see movement.



Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \quad P(X) = \sum_Y P(X \mid Y)P(Y)$$

Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{0.75 \cdot 0.4 + 0.75 \cdot 0.6} = 0.4$$

Here the fruit told us **nothing** about the box — both boxes have the same orange fraction. Evidence is only informative when the hypotheses predict different things.

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. The rest of this chapter makes them precise and extends them beyond counting.

Signpost explicitly: everything just used informally gets a definition in the coming modules. This unit is the map, not the territory.



Module 1

Events and the Algebra of Sets

Guiding Question: *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Speaker notes

Note what the definition does *not* say: nothing about the world being indeterministic. Ask students for a "natural random experiment" — rainfall, click-through, a queue — where nobody designed the configuration.



Randomness as Missing Information

A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.

Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Speaker notes

This is the epistemic-modeling principle in probability form (P3): calling something random is a statement about our knowledge, not only about the system. Good place to ask where that matters in data science — noise vs. unmodelled signal is the same distinction.



Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event** the event A **occurs** iff the observed outcome ω is a member of A . The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set we return to this when continuous distributions appear.

Math first (P5). The load-bearing word is "exactly one" — outcomes must be mutually exclusive and exhaustive, which is a modelling decision, not a mathematical one. The outcome/event distinction is the one students blur.

Important Event Types

$$\emptyset$$

impossible event

$$|A| = 1$$

elementary event

$$\Omega$$

certain event

$$A \subseteq B$$

A is a sub-event of B — " A entails B "

$$\overline{A} := \Omega \setminus A$$

complementary event

$$A \cap B = \emptyset$$

A and B are incompatible

2x3 grid, read as vocabulary rather than results. "Entails" is worth dwelling on: set inclusion is logical implication between events, which is why the next unit can do algebra with pictures.

Examples

One coin flip

$$\Omega = \{H, T\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

Event "heads": $E = \{H\}$

Two coin flips

$$\Omega = \{HH, HT, TH, TT\}$$

Event "at least one head": $E = \{HH, HT, TH\}$

Four outcomes, sixteen possible events.

Have students write down the event "exactly one head" ($\{HT, TH\}$) and the event "the first flip is heads" ($\{HH, HT\}$) — same sample space, different questions.

The Event Space Grows Fast

Three-sided die: $\Omega = \{1, 2, 3\}$, so $\mathcal{P}(\Omega)$ has $2^3 = 8$ events, ordered by inclusion:

$\{1, 2, 3\}$ $\{1, 2\}$ $\{1, 3\}$ $\{2, 3\}$ $\{1\}$ $\{2\}$ $\{3\}$ $\emptyset$

For $|\Omega| = n$ the event space has 2^n elements — which is exactly why the next unit is about counting.

The lattice diagram of the original is replaced by three inclusion levels; the point is the 2^n growth, not the drawing. This motivates counting and, later, why we cannot always take the full power set for infinite Ω .

Not Every Sample Space Is Finite

Dosage of a drug

$$\Omega = (0, \infty)$$

Uncountably many outcomes.

Race with 7 teams

$$\Omega = \{\text{all orderings of } 1, \dots, 7\}$$

$7! = 5040$ outcomes.

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Speaker notes

Close the module on the modelling move (P3). Ask what is lost by modelling dosage as a positive real (measurement precision, dosing intervals) — the same question returns for every feature in every dataset later in the course.



Venn Diagrams

Venn diagrams are the graphical representation of events:

- the rectangle represents the sample space Ω
- circles represent events $A \subseteq \Omega, A \in \mathcal{P}(\Omega)$

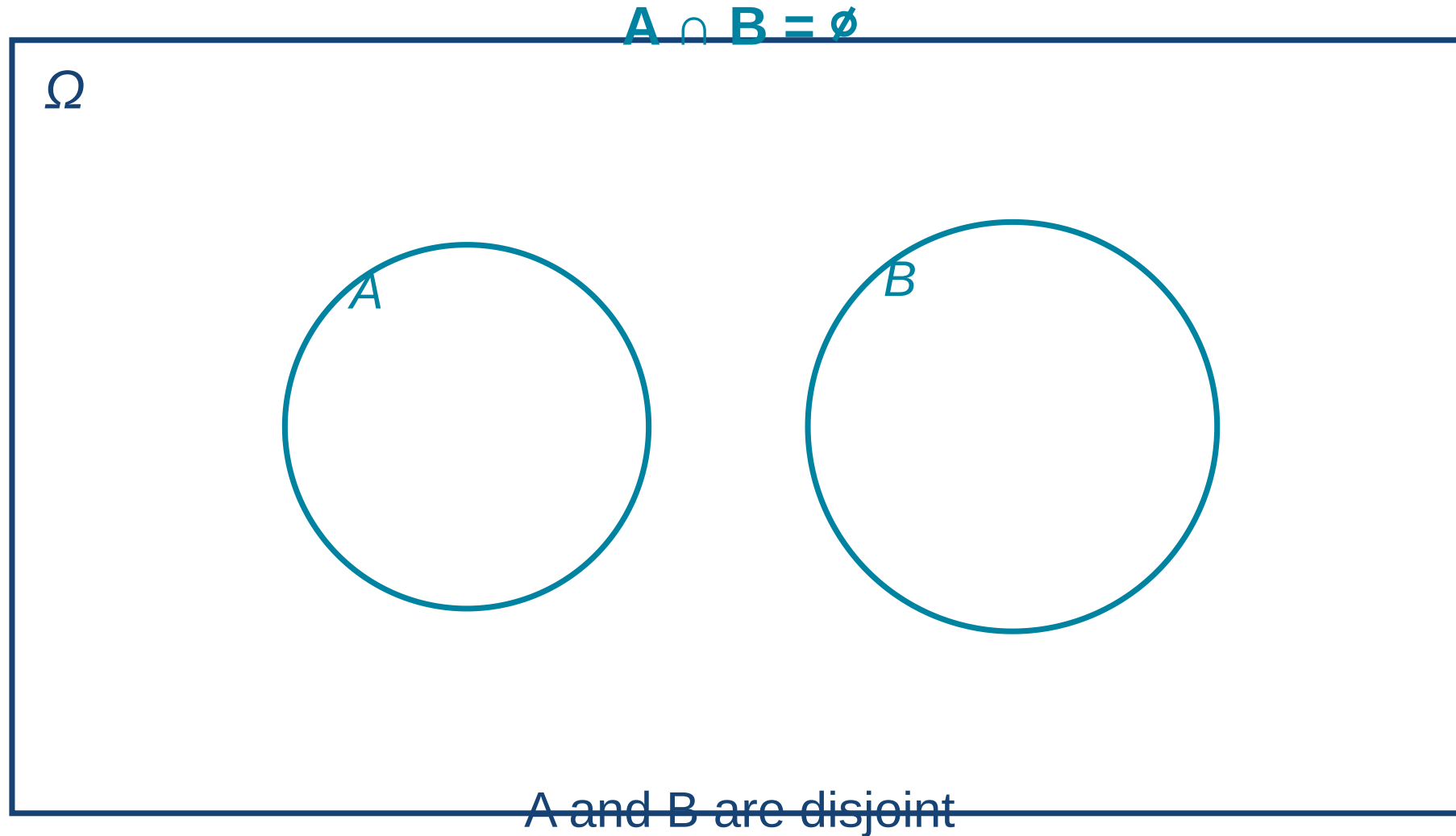
Every region of the picture *is* a set — and every set operation is a region you can point at.

Bridge from sets to pictures before any formula (P4). The payoff of the diagram: probability laws become area arguments in the next unit.



Important Event Types

Before each step: which region will light up?



Speaker notes

Stepped visualization (P6, tier 2), one click per event type: the event A ; the complement $\bar{A} = \Omega \setminus A$ the subset $B \subset A$ disjoint events $A \cap B = \emptyset$. Predict-then-check on every click — students name the region first, the highlight confirms. Disjointness is the one to slow down on: it is a statement about events, not about probabilities.

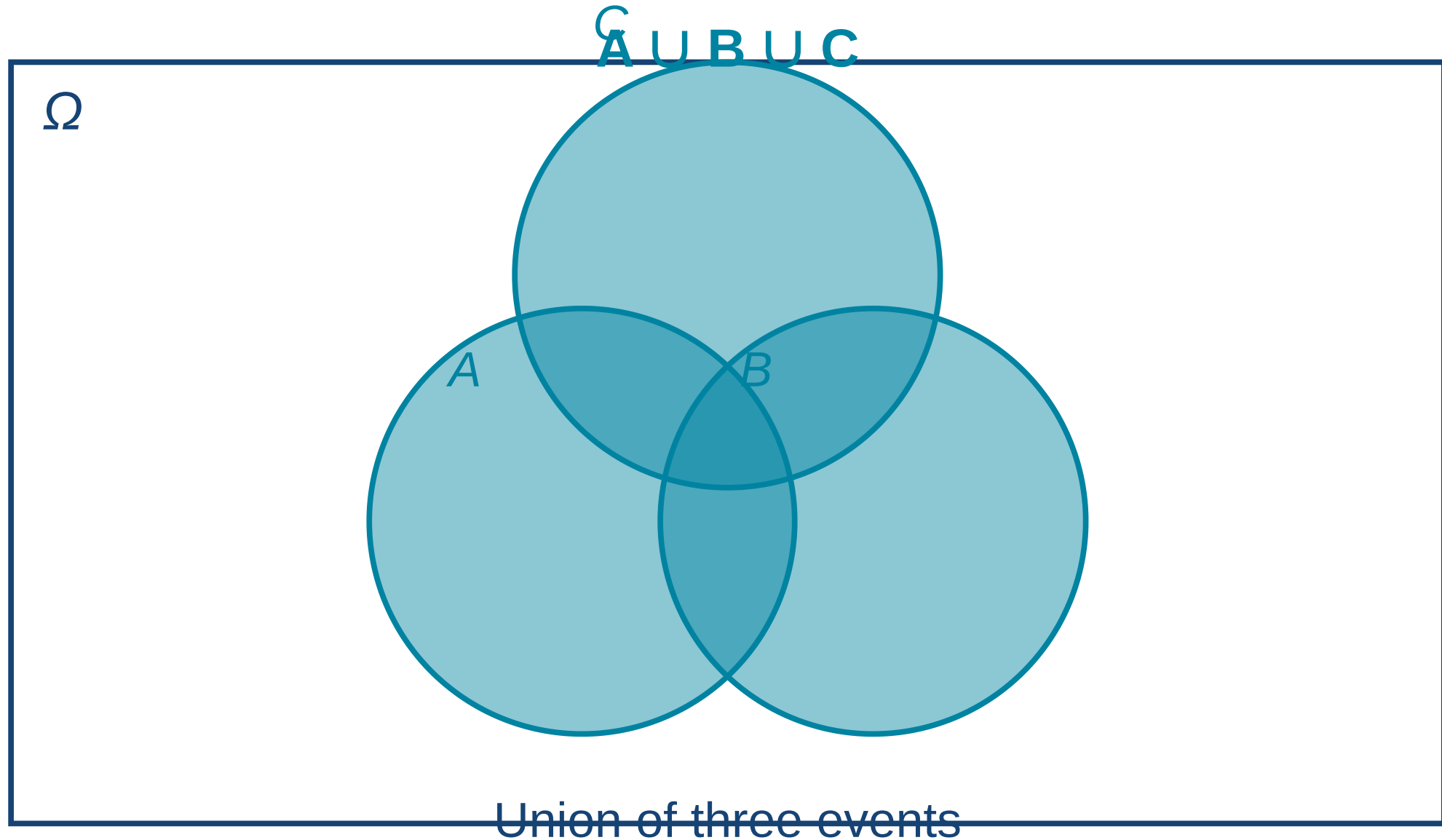


Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions



Second stepped pass with three sets: $A \cup B$, $A \cap B$, $A \cap B \cap C$, $A \cup B \cup C$. Same predict-then-check rhythm.



Mini-Exercise

Sketch (pen and paper) the regions for:

1. $(A \cup B) \cap C$
2. $(A \cap B) \cup C$

Are they the same region? Which algebra law says so — or says they are not?

Mini-Exercise — Solution

Solution fragmented per the animation policy (solutions always click-gated). This replaces the last two TikZ slides of the original unit with an active task — same content, students draw instead of watch.

Mini-Exercise — Solution

They differ: $(A \cup B) \cap C$ lies entirely **inside** C , while $(A \cap B) \cup C$ contains **all** of C plus the lens $A \cap B$ outside it.

The distributive laws expand each into a different expression — the algebra distinguishes exactly what the picture shows.

Module 2

Counting

Guiding Question: How do we determine the size of a sample space we could never write down?

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Motivate combinatorics as the arithmetic of sample spaces, not as a separate topic. Flag the assumption doing the work here — "equally likely" — which the next unit replaces with an axiom system that does not need it.



Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175,760,000$$

The phrase "regardless of the earlier stages" is the hidden assumption — it fails as soon as choices interact (no repeated letters, say), which is exactly what the four cases below systematise.

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation**
(rankings)
does not → **combination**
(committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange
all n ?

Give students a problem and have them answer these three questions *before* reaching for a formula. Nearly every counting error is a misread of one of these three, not an algebra slip.

The Four Basic Cases

Ordered, without repetition

k -permutation of n :

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k -combination of n :

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

2x2 grid — the whole unit in one slide. Keep it visible while working examples; students should learn to locate a problem in this square rather than recall four formulas in isolation.

Permutations

Definition (k -permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n - 1$, ..., down to $n - k + 1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3,628,800$.

The stage-by-stage argument *is* the basic counting principle applied with a shrinking pool — worth making explicit so the formula is derived, not given.

Combinations

Each k -combination corresponds to $k!$ different k -permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.

Remember it as *over-count, then correct*.

Speaker notes

The over-count-then-divide pattern recurs constantly. Ask why we divide by $k!$ and not by something else — because every unordered selection was counted exactly $k!$ times, uniformly.



Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n - k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i : groups **containing** i , plus groups **not containing** i .

Binomial theorem.

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Speaker notes

Both identities are proved by counting the same set two ways — the standard combinatorial proof technique. The binomial theorem's proof is the same idea: expanding the product gives 2^n terms, and $x^k y^{(n-k)}$ appears exactly as often as there are ways to pick which k factors contribute x .



Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

Alternative intuition. Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the previous unit.

Speaker notes

Close the loop with the sample-space unit: the 2^n we asserted there is now derived twice. Ask which of the two arguments they would trust more under pressure — the binary-choice one generalises better.



Mini-Exercise

A password consists of 4 characters from a 26-letter alphabet.

1. How many passwords are there?
2. How many with **no repeated letter**?
3. How many **sets** of 4 distinct letters (order ignored)?

Mini-Exercise — Solution

Three of the four cases in one exercise ask which case is missing (unordered with repetition) and to invent a password-flavoured question for it.



Mini-Exercise — Solution

1. Ordered, with repetition

$$26^4 = 456,976$$

2. Ordered, no repetition

$$\frac{26!}{22!} = 358,800$$

3. Unordered, no repetition

$$\binom{26}{4} = 14,950$$

Takeaway

Counting reduces to three questions — **order?** **repetition?** **how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

Module 3

What Is a Probability?

Guiding Question: *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A , assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work. Assumption 2 is the one we will have to give up.

Intuition first (P4), and it connects directly to the Venn unit: events were regions, now regions have size. Ask what happens to the picture if some parts of Ω are "heavier" than others — that is precisely what the axioms will allow.

Two Ways to Read a Probability

Frequentist

The probability of an elementary event a is the fraction of repetitions of the experiment that yield a .

Requires the experiment to be repeatable.

Bayesian

The probability of a is the degree of certainty that a will happen — even without conducting the experiment.

Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

Interpretation over recitation (P2): ask which reading makes sense for "the probability that this patient has the disease" (no repetition available) versus "the probability that this machine part fails within a year". The distinction returns whenever we call something a prior.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment** its probabilities need not be estimated, they are implied by the setup.

This is the definition the counting unit was preparing. Emphasise that it is only usable under the Laplace assumption when that fails, probabilities must come from data.

Why That Definition Is Not Enough

The circularity. "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

Speaker notes

This slide is the pivot of the module and answers the guiding question. Let the tension sit for a moment before the axioms arrive: students should feel why an axiomatic move is the honest way out, not a formal ritual.



The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \rightarrow \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$

$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a σ -**algebra** of measurable sets, and Axiom III is required for **countably many** disjoint

Speaker notes

Math first (P5). Ask: which axiom does the area picture make obvious? (All three — areas are non-negative, the whole rectangle is everything, disjoint regions have additive areas.) The axioms are the area intuition, minus the requirement that the areas be equal.



Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

Whenever a later theorem opens with "let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space", students should hear "we have agreed on outcomes, questions, and a measure".



What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, II)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive $A_1, \dots, A_p$: $P(A_1 \cup \dots \cup A_p) = P(A_1) + \dots + P(A_p)$.

Speaker notes

2x2 grid of consequences. Have students justify inclusion–exclusion on the Venn picture — the overlap was counted twice. Note that "A entails B implies A is no more likely than B" is monotonicity: the logic of events transfers to numbers.



Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Stabilise the epistemic move: an axiomatic definition says what the object must do, leaving where the numbers come from to the modeller. That freedom is what lets the same theory carry frequentist and Bayesian readings.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

The formula is trivial; the modelling is not. Every error in the examples below comes from an Ω where outcomes are not equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs?
Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

The classic trap. If a student proposes the 21 unordered pairs, ask them to bet on "double six" versus "one and six" — the intuition corrects itself fast.



Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs?
Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$$|\Omega| = 11 \cdot 10 \cdot 9 = 990$$

Favourable: white-black-black, black-white-black, black-black-white = $6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$$|\Omega| = \binom{11}{3} = 165$$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

Speaker notes

Same answer, two sample spaces — the point of the slide. Both are legitimate *because* both are equiprobable: ordered draws are equally likely, and so are unordered groups. Mixing the two (favourable counted unordered, total counted ordered) is the error to warn against.



The General Rule

When selecting k items from n , you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Connects straight back to the four cases in the counting unit: the choice of case is a modelling choice about Ω , not a property of the problem.

Mini-Exercise

Five people are chosen from **10 couples** (20 individuals).

What is the probability that the five chosen are **mutually unrelated** — no couple is fully represented?

Mini-Exercise — Solution

Solution fragmented. The structural insight: "choose the groups, then choose within each group" is the multiplication principle applied to combinations — a pattern that reappears in stratified sampling and in naive Bayes.

Mini-Exercise — Solution

Choose 5 of the 10 couples, then one person from each:

$$P(N) = \frac{\binom{10}{5} \cdot 2^5}{\binom{20}{5}} = \frac{252 \cdot 32}{15504} \approx 0.52$$

Order is ignored consistently in numerator and denominator.

Module 4

Conditioning, Independence, Bayes

***Guiding Question:** A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

Intuition first (P4): conditioning is not a new kind of probability, it is the old probability computed in a **smaller sample space**. Let students do the count before the definition appears.

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1, 1), (1, 2), \dots, (6, 6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3, 1), \dots, (3, 6)\}$ remains
- Outcomes like $(2, 4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3, 5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability).

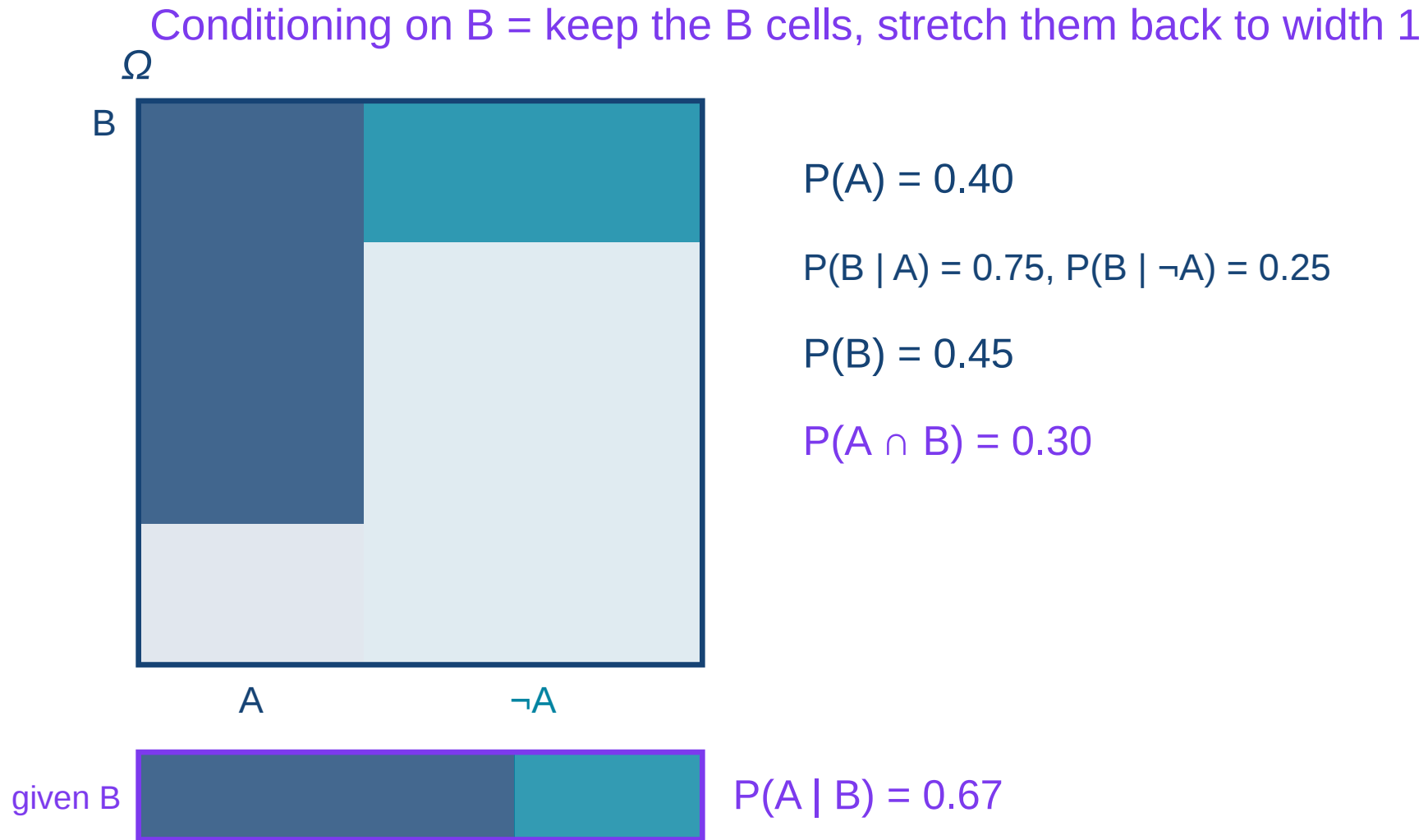
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B .

Back to the stone: *given that the stone landed in area B , how likely is it that it also landed in A ?*
We renormalise by the area of B , because B is now the whole world.

Math first (P5) after the dice intuition. The division by $P(B)$ is exactly the renormalisation students performed implicitly when they used the six remaining outcomes.

Conditioning, Drawn



Before the last click: what must the two shaded cells be divided by?

Speaker notes

Stepped visualization (P6): the A columns; B's share inside each column; keep only B; renormalise. The stretch factor *is* the $1/P(B)$ in the definition, so the picture performs the formula rather than illustrating it.

Then use the sliders. Two settings are worth doing live: drag $P(B|A)$ and $P(B|\neg A)$ to the same value — the bar's split stops moving and $P(A|B) = P(A)$, which is independence, met two units early and by accident. Then set $P(B|\neg A)$ near zero: B becomes nearly conclusive evidence for A. Ask what has to be true of the two conditionals for evidence to be worth anything.



Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$$F = \{(h, h), (h, t)\}$$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$$A = \{(h, h), (h, t), (t, h)\}$$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B , two different pieces of information, two different answers. **Why is (b) smaller?**

Speaker notes

Force interpretation (P2). The information in (a) is stronger — it names *which* coin — so it eliminates more of the sample space. This is the classic "boy or girl" puzzle in disguise expect disagreement, which is the point.



Total Probability

Theorem (Total probability). Let $B_1, \dots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \dots \cup B_k$ and $P(B_i) > 0$. Then for any event A :

$$P(A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \dots \cup B_k) \cap A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) \cdot P(A \mid B_i)$$

Two axioms do all the work: completeness of the partition, then exclusiveness for additivity. This theorem is the denominator of Bayes' rule — say so now, so that Bayes looks assembled rather than conjured.

Rules That Follow

Product rule

$$\begin{aligned}P(A \cap B) &= P(B) \cdot P(A \mid B) \\ &= P(A) \cdot P(B \mid A)\end{aligned}$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B , the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B .

2x2 grid. The top-right box is already Bayes' theorem — flag it, then move on; the next unit gives it its name and its interpretation. The bottom two boxes are the most common exam mistake: you may complement the *event*, not the *condition*.

Chain Rule

$$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \dots \cap E_{n-1})$$

By successive application of the product rule.

Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Do not belabour the proof; the payoff is the remark. The chain rule plus conditional independence is the entire idea behind belief networks.

Takeaway

Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \quad \text{and} \quad P(A \mid \overline{B}) = P(A)$$

Independence means: **learning B changes nothing about A .**

The conditional form is the meaningful one; the product form is the one that is easy to check and stays defined when $P(B) = 0$. Emphasise the reading: knowing B tells you nothing new.

The Classic Confusion

Statistical independence does not mean the events are disjoint.

Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Speaker notes

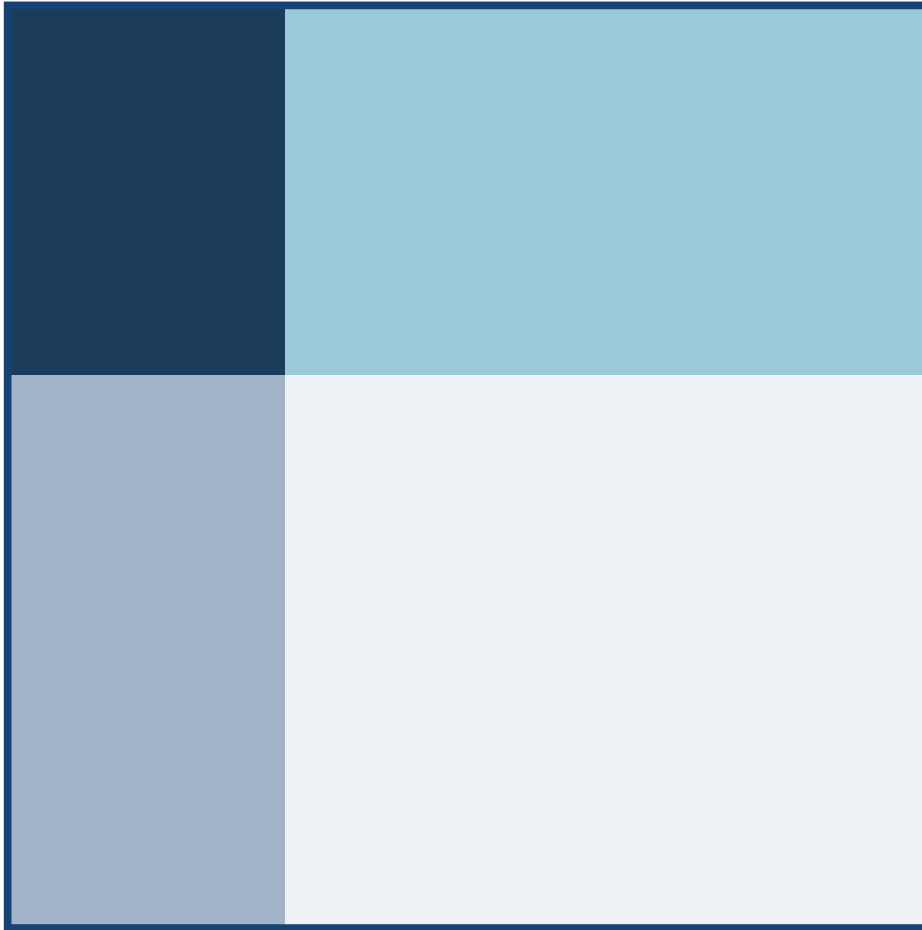
Ask students to state, in one sentence, why disjointness is the opposite of independence. The Venn picture misleads here — non-overlapping circles look "unrelated", which is exactly wrong.



Independence Is About Areas, Not Overlap

$$0.3 \cdot 0.4 = 0.12 = P(A \cap B) \rightarrow \text{independent}$$

Ω



$$P(A) = 0.3$$

$$P(B) = 0.4$$

$$P(A \cap B) = 0.12$$

Speaker notes

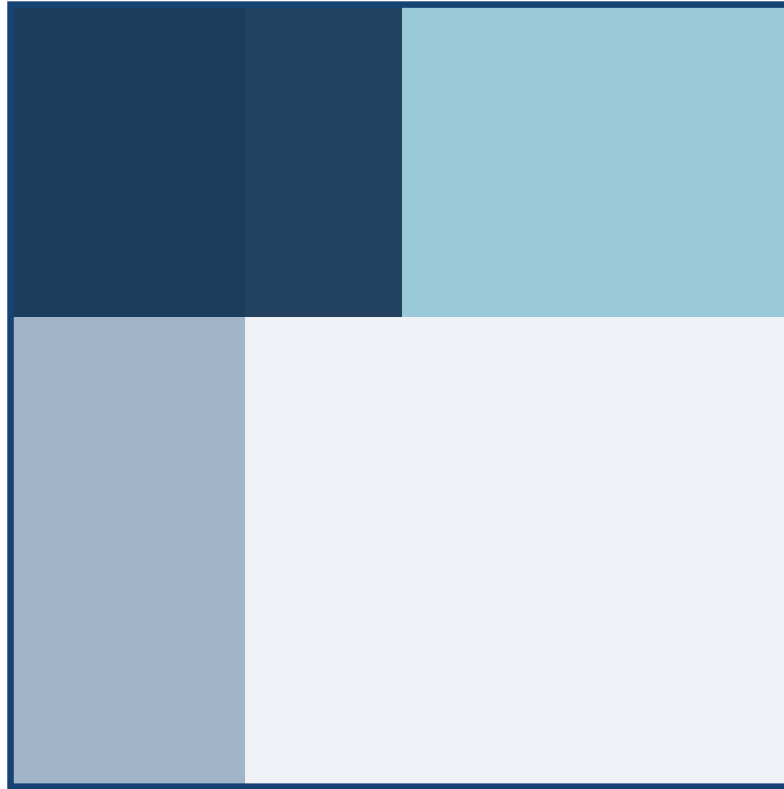
Stepped viz (P6): band A, band B, their intersection, then the test. $0.3 \cdot 0.4 = 0.12$ — independent. Ask before the last click whether they expect equality. The lesson: independence is a numerical coincidence of areas, not a visual property of the drawing.



The Same Picture, Dependent

$$0.3 \cdot 0.4 = 0.12 \neq 0.2 = P(A \cap C) \rightarrow \text{dependent}$$

Ω



$$P(A) = 0.3$$

$$P(C) = 0.4$$

$$P(A \cap C) = 0.2$$

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

Speaker notes

Same marginals as the previous slide, different joint. This is the payoff of the pair: the marginals never tell you whether events are independent. Later, that is exactly why joint distributions are needed.



More Than Two Events

Events $A_1, \dots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \dots, i_l$:

$$P(A_{i_1} \cap \dots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Give the standard counterexample if time allows (two fair coins and the event "the results agree" are pairwise independent but not mutually independent). The distinction matters for naive Bayes, which assumes far more than pairwise.

Takeaway

Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given C** (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A .**

Math first (P5), but immediately give the reading. The second form is the one used in practice: it says B becomes redundant in the presence of C.

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$$\frac{1}{16} \neq \frac{3}{32}, \text{ so } A \not\perp B$$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

$$\text{so } A \perp B \mid C$$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

The whole idea in one comparison. Conditioning on C *created* independence that did not exist before. Ask for a real example: ice-cream sales and drownings are dependent, but conditionally independent given the temperature.

Why This Matters

Explaining away

A common cause C can make two effects look related.

Model size

Conditional independence lets a joint distribution factorise into small pieces.

Later in the course

Naive Bayes assumes it.
Graphical models are built from it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Flag forward links without using them: students only need to know that this definition is the hinge for models later. The warning box is worth a moment — students routinely assume one form gives the other.

Takeaway

Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Intuition-first entry (P4). Collect guesses by show of hands *before* clicking: most say 99%. The surprise is the motivation for the whole module. Do not explain yet — the next slide lets them count it themselves.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

$$P(\text{sick} \mid \text{positive}) = 1/11 \approx 9\%$$



Speaker notes

Stepped visualization (P6 tier 2), one click per step: the population the one sick person the ~10 false alarms among the 999 healthy restricting to the 11 positives. Ask before the last click: "of these 11 flagged people, how many are sick?" Everything here is counting and restriction of the sample space — no new machinery. This *is* conditioning, drawn.



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Name the confusion explicitly — swapping the two conditionals is the single most common probabilistic error, in exams and in the newspapers. Bayes' theorem is the machine that converts one into the other.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

posterior — what we believe after prior — belief before the evidence
likelihood evidence how well A explains B how well B explains A

Which of the four was "99% accurate", and which one did the intuition ignore?

Math-first-then-meaning (P5), legitimate here because the intuition already exists. Stepped equation walk-through: each click highlights one part and its brace — posterior, likelihood, prior, evidence — while the rest dims. Map the disease test onto the symbols as you go: A = sick, B = positive test. $P(A) = 1/1000$ is the term everyone forgets. Reference: (Sheldon & others 2002).

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Speaker notes

One line: expand, multiply by $P(A)/P(A)$, recognise the likelihood. Worth asking which step needed $P(B) > 0$ and which needed $P(A) > 0$.



Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \dots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i | B) = \frac{P(B | A_i) \cdot P(A_i)}{P(B)} = \frac{P(B | A_i) \cdot P(A_i)}{\sum_{j=1}^k P(B | A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Speaker notes

The hypotheses partition Ω — "sick vs. healthy" was the $k = 2$ case. This is the form that reappears as a classifier (naive Bayes) and in belief networks: score every hypothesis by likelihood $\times$ prior, then normalise.



Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^k P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Solution fragmented per the animation policy. The punchline is the comparison with 9%: identical likelihoods, different prior, completely different clinical meaning.

Mini-Exercise — Solution

Out of 1,000: 50 sick $\rightarrow \approx 50$ true positives 950 healthy $\rightarrow \approx 10$ false alarms.

$$P(\text{sick} \mid \text{positive}) \approx \frac{50}{60} \approx 83$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior** $\propto$ **likelihood** $\times$ **prior**.
Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Stabilise both halves: the base-rate trap (9%) and the prior's power (83%). Both return in Bayesian decision making and graphical models.

Module 5

Random Variables and Distributions

Guiding Question: *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y :

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

Intuition first (P4). Ask what was lost in the compression (which coin showed heads) and what was gained (a quantity we can average, compare, plot). That trade is the entire point of a random variable.

Definition

Definition (Random variable). A random variable X maps outcomes to a measurable range E : $X : \Omega \rightarrow E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

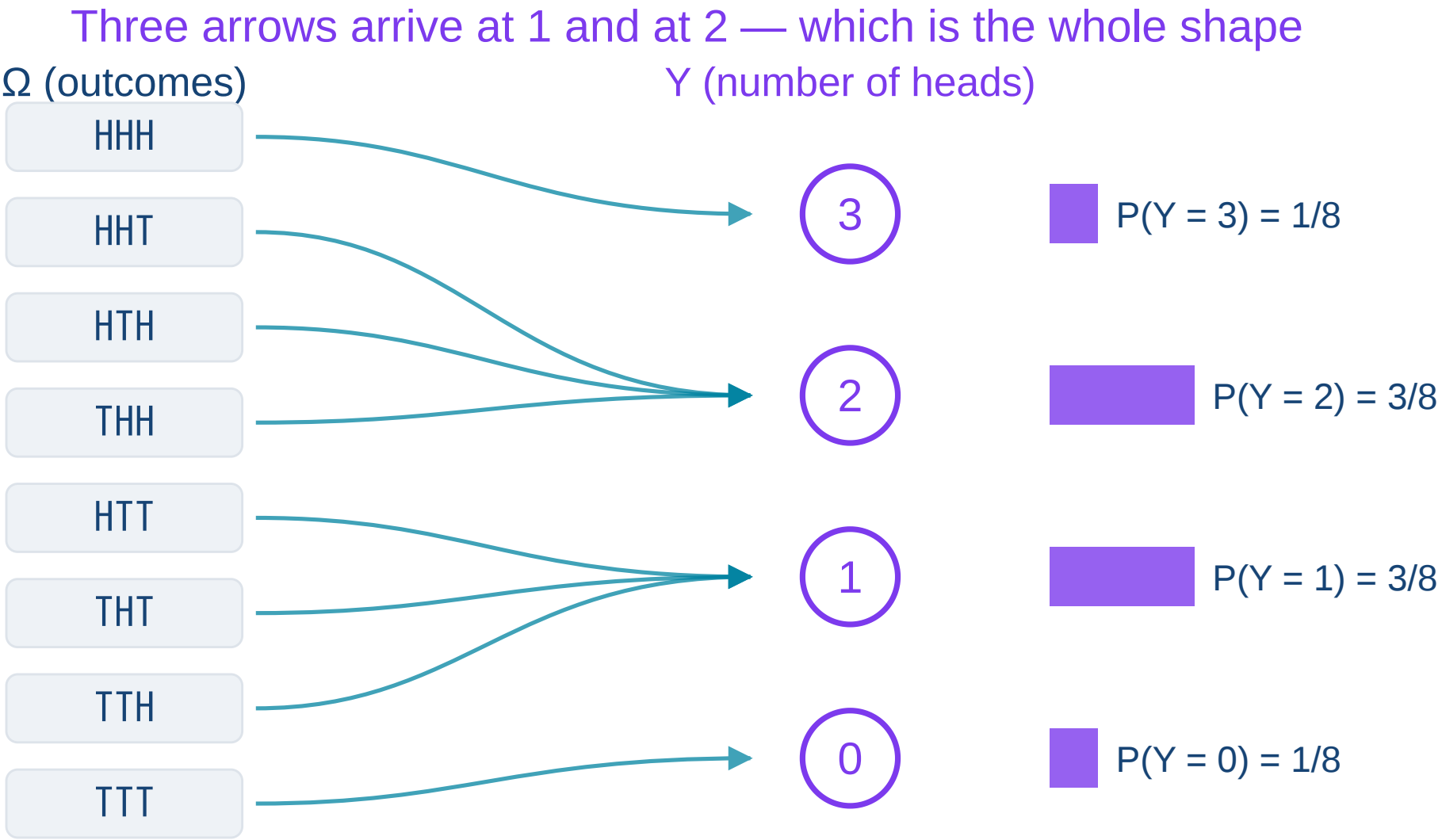
A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω , not in X .

Speaker notes

The naming trap is worth one minute; it removes a persistent confusion. The formula says: to get the probability of a value, collect all outcomes that map to it and use the measure we already have.



The Function, Drawn



Stepped visualization (P6), three coins: the eight outcomes; the arrows, one per outcome; the four values with their counts; the resulting mass function. Two things to say out loud while stepping. First, *every* outcome has exactly one arrow leaving it — that is what makes Y a function. Second, three arrows arrive at 1 and three at 2, and that is the entire reason those values are more probable. The probability was never invented; it was inherited.

Examples

Two coins

$$\Omega = \{HH, HT, TH, TT\}$$

X = number of heads

People

$$\Omega = \{\text{person}_1, \text{person}_2, \dots\}$$

X = height of the person

Two dice

$$\Omega = \{\langle n_1, n_2 \rangle\}$$

$$X(\omega) = n_1 + n_2$$

The middle example is the one that connects to data science: a dataset column *is* a random variable — a function from records to values. Say it explicitly; it makes the next chapters feel continuous with this one.

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — the subject of the next two units.

Keep this light: the deep measure-theoretic reasons are out of scope. What students need is the practical consequence — sums for discrete, integrals for continuous.

A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1 - p)p, \quad P(X = 3) = (1 - p)^2 p$$

$$P(X = n) = (1 - p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Speaker notes

This is the geometric distribution, met before it is named — fine, because it is constructed from the chain rule the students already have. Ask them to verify the sum is 1 (geometric series) if the group is comfortable with it.



Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X , the **probability mass function** is

$$p(a) = P(X = a)$$

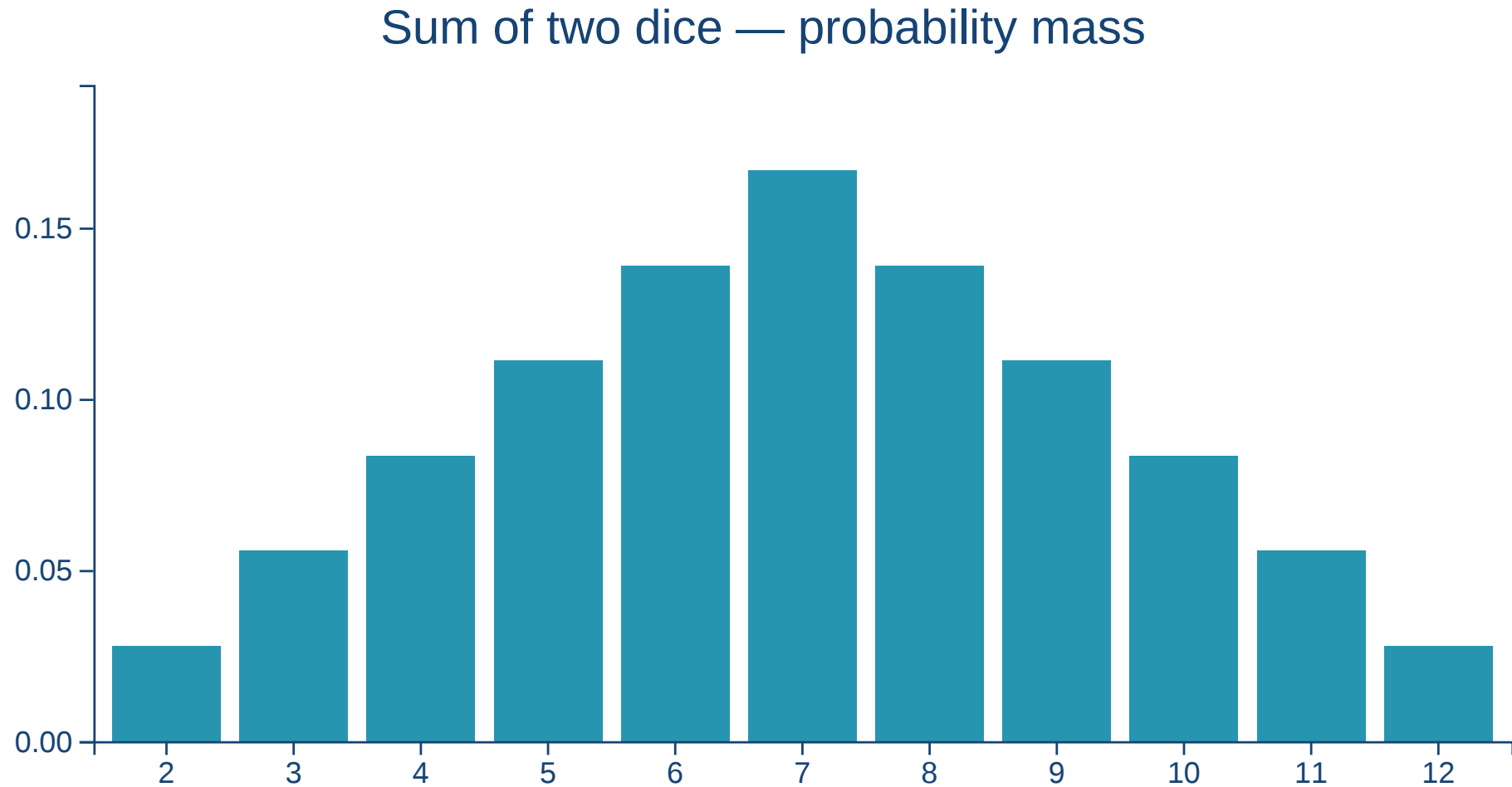
$p(a) > 0$ for at most countably
many a

$$\sum_i p(a_i) = 1$$

p is itself a probability measure

The PMF is just the probability measure, re-indexed by values instead of outcomes. Everything from the axioms still applies — which is why the three properties are consequences, not new assumptions.

Example: Sum of Two Dice



The triangular shape is the counting from earlier units, drawn: six of the 36 outcomes give 7, only one gives 2.

Speaker notes

Connect the picture to the Laplace count they already did. Ask why the shape is symmetric and peaked — more ways to make a middling sum. This is the first hint of why sums of random quantities concentrate.



Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$ the first *central* moment $E[X - E[X]]$ is always 0

For a real function g :

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b :

$$E[aX + b] = aE[X] + b$$

Speaker notes

The linearity identity is the one used constantly later (in regression, in evaluation). Note that $E[g(X)] \neq g(E[X])$ in general — ask for a counterexample with $g(x) = x^2$.



Variance

Definition (Variance).

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\sigma = \sqrt{\text{Var}(X)}$ is the **standard deviation**, in the same unit as X

Speaker notes

Two things to stabilise: the computational identity $E[X^2] - E[X]^2$, and why we report σ rather than the variance (units). The b disappearing under variance is worth a question: shifting data does not change its spread.



Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

Binomial $B(n, p)$ — n independent trials

$$p(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

The binomial PMF is the counting formula times the probability of one specific sequence — combinatorics and independence in a single line.

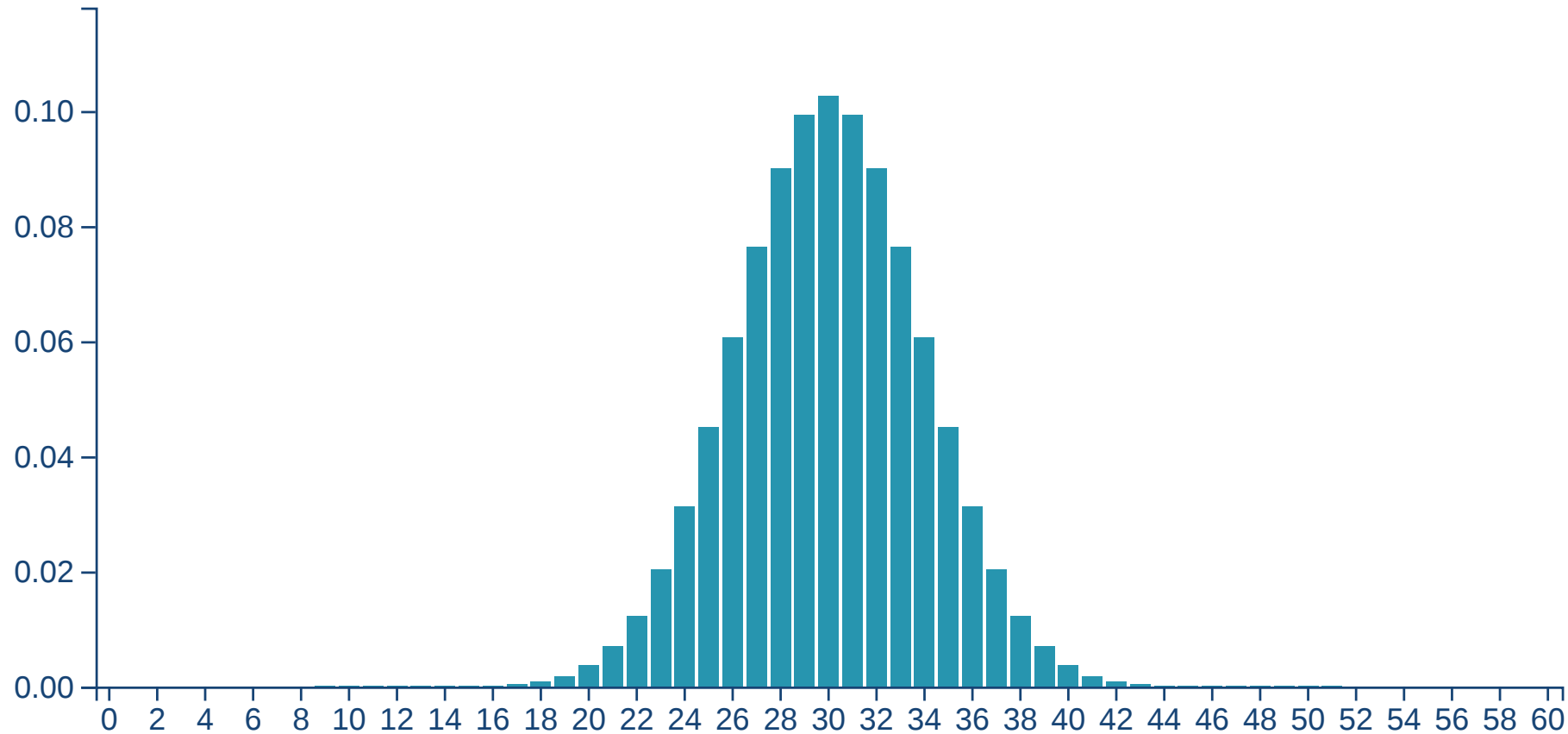
Speaker notes

Point at where each earlier unit shows up: the binomial coefficient from counting, the product $p^k(1-p)^{n-k}$ from independence. Nothing here is new machinery, which is exactly the message.



How the Shape Depends on the Parameters

Binomial $B(n = 60, p = 0.5)$ — probability mass



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Predict-then-check (P6). Two things happen at once: the spread in absolute terms grows ($\sqrt{n}$) while the shape becomes symmetric and bell-like. The second observation is the central limit theorem arriving early — name it only as an observation here the normal distribution is defined in the next unit.

Mini-Exercise

A quiz has 10 questions with 4 options each. A student guesses every answer.

1. Which distribution describes the number of correct answers?
2. What is the expected number correct?
3. Is passing (≥ 5 correct) likely?

Mini-Exercise — Solution

Part 3 rewards recognising that the answer needs a sum over the tail, not one PMF value. The modelling caveat matters more than the arithmetic.

Mini-Exercise — Solution

1. $B(10, 0.25)$ — ten independent trials, constant success probability
2. $E[X] = 10 \cdot 0.25 = 2.5$
3. $P(X \geq 5) \approx 0.078$ — under 8%

The independence assumption is what makes this a binomial ask when it would fail in a real exam.

Takeaway

A discrete distribution is a PMF expectation and variance compress it into **where it sits** and **how much it spreads**. Bernoulli and binomial are the counting units in probabilistic clothing.

From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \quad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a .

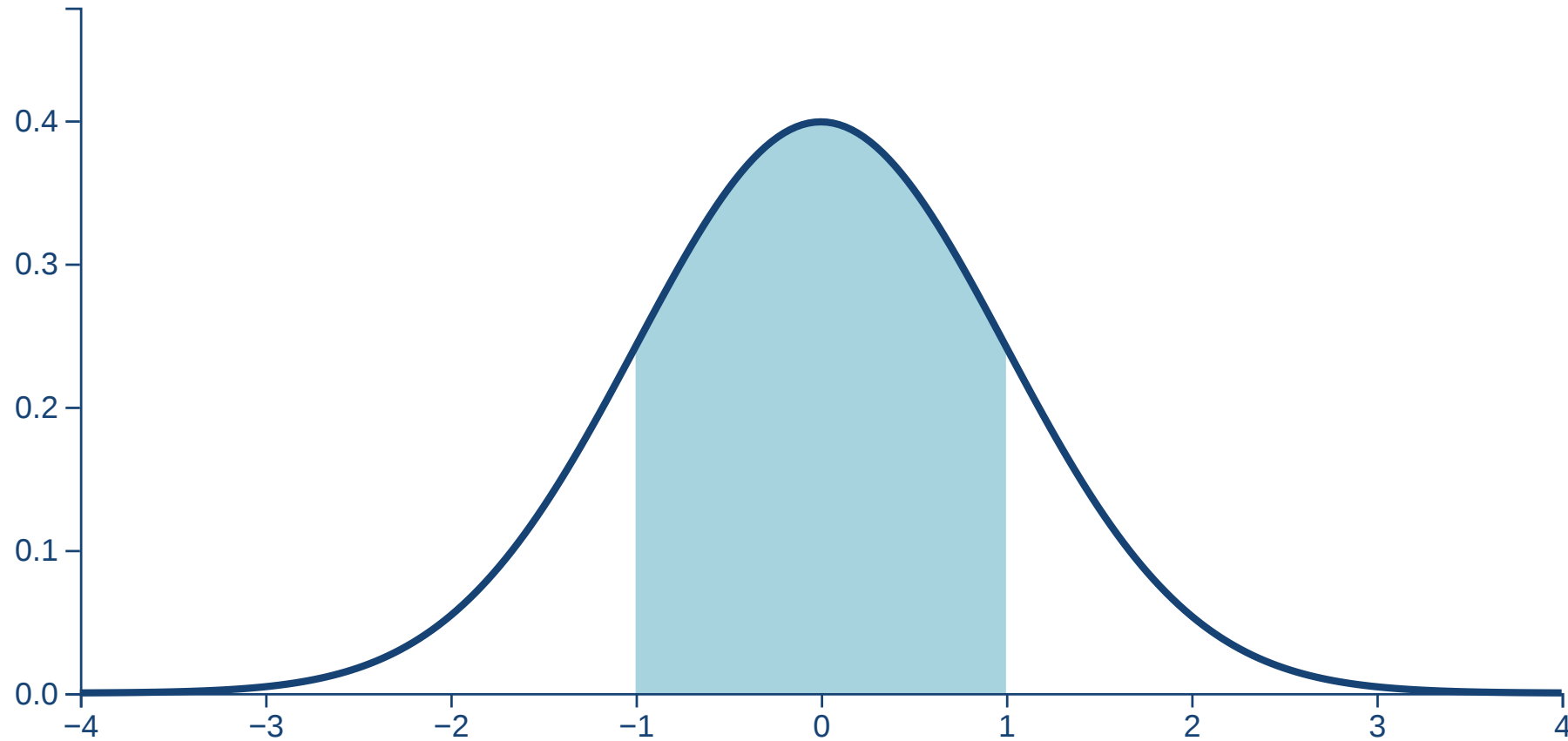
Speaker notes

The density-is-not-probability point is the single biggest source of confusion in this unit. A useful question: what is the probability that a person is exactly 180.000... cm tall? Zero — probability lives on intervals now.



Probability Is Area Under the Curve

Normal $N(\mu = 0, \sigma = 1)$ — density



The shaded region is $P(-1 \leq X \leq 1)$ — the same "probability as area" intuition from the axioms of probability, now with a curve instead of a Venn diagram.

Deliberate callback (P7 spiral): area was the intuition for the axioms, and it still is. Ask what the total area must be, and why that forces the normalisation constant in the density formula.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\text{Var}(X) = E[(X - \mu)^2]$.

Emphasise the continuity of concepts across the discrete/continuous divide: only the summation symbol changed. Students who see this stop treating the two cases as separate theories.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

Every interval of equal length is equally likely — the Laplace assumption, continuous version.

Normal $N(\mu, \sigma^2)$

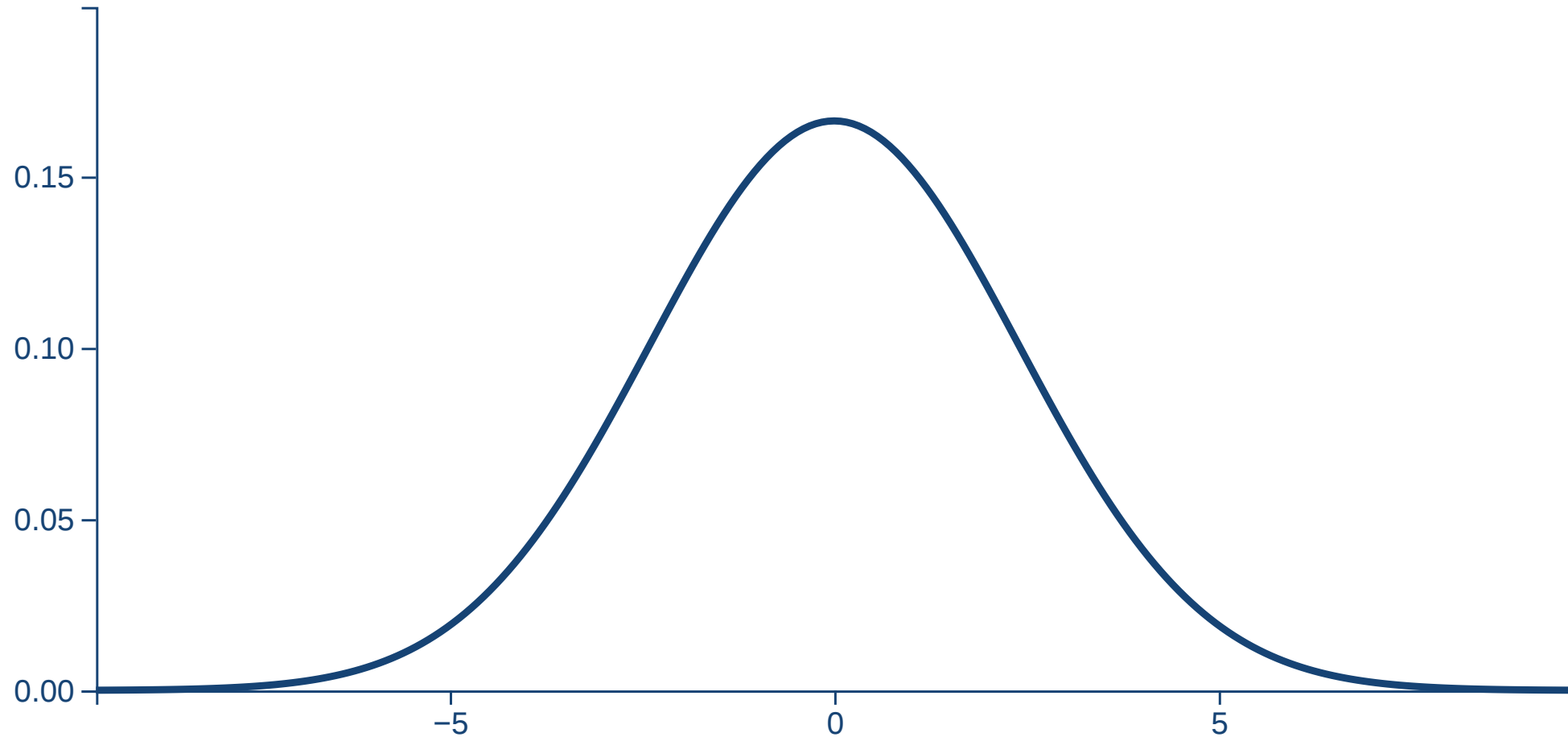
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\text{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

The uniform is worth naming as the continuous Laplace assumption — it makes the earlier discussion feel like one idea rather than two.

What σ Does to the Shape

Normal $N(\mu = 0, \sigma = 2.4)$ — density



The peak falls as the curve widens — the area is always 1.

Predict-then-check: ask what must happen to the height when the width grows. The constraint "total area = 1" is doing the work, and it is the same normalisation axiom from the beginning of the module.

Why the Normal Is Everywhere

The **central limit theorem**: under mild conditions, averages of many independent observations converge to a normal distribution — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown, and why the binomial started looking bell-shaped as n grew in the previous unit.

State the CLT, do not prove it. The callback to the binomial sweep is the point: students already saw it happen. This also sets up sampling and hypothesis testing at the end of the module.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values confined to a finite interval $[a, b]$
- **Dirichlet** — distributions over probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

The last two (Beta, Dirichlet) are named for recognition only — they return in the probabilistic machine learning course as priors over probabilities. That is a legitimate forward *pointer*, not a forward dependency: nothing here uses them.

More Than One Random Variable

Univariate

one random variable

Bivariate

two random variables

Multivariate

three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Make the dataset connection explicit: a table with columns age, income, and churn is a sample from a trivariate joint distribution. Everything in the rest of the course operates on joints.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b) da db$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\dots)$ for the probability that $\{X_1, X_2, X_3, \dots\}$ is in state $a, b, c, \dots$

Flag the convention as a deliberate simplification, and say why it is safe: the rules (marginalise, condition, factorise) are identical only the mechanics of summing versus integrating differ.

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \dots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \dots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Marginalisation is the total-probability theorem, applied to random variables — say so, since students met it three units ago. That continuity is worth more than the formula.



Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Speaker notes

Have students read one cell aloud as a sentence ("7.5% of families have one boy and two girls"). Then ask whether B and G are independent — they are not, since the family size couples them. Good preparation for covariance.



Covariance

$$\text{cov}(A, B) = E[(A - E[A])(B - E[B])]$$

$$\text{cov} > 0$$

A large when B large —
positively related

$$\text{cov} < 0$$

A large when B small —
negatively related

$$\text{cov}(A, A) = \text{Var}(A)$$

variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

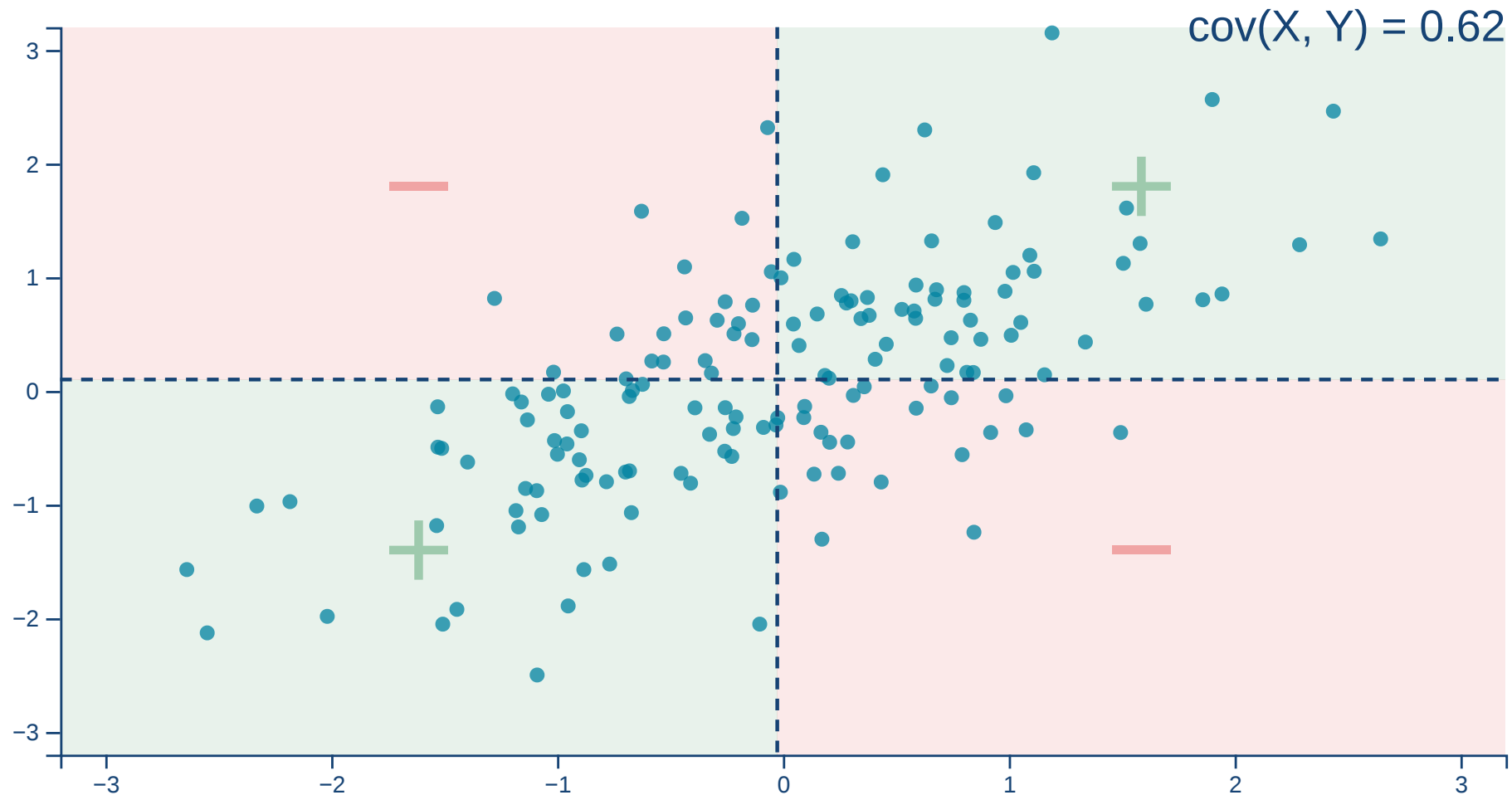
Speaker notes

The warning is the load-bearing statement. A classic counterexample: X uniform on $[-1,1]$ and $Y = X^2$ are dependent but have zero covariance. This is why later methods that rely on correlation can miss real structure.



Covariance Sees Only One Kind of Structure

Slide ρ , then switch the relation. Predict the covariance first.



Predict first: with $Y = X^2 - 1$, every point is fixed by its x . The covariance?

Speaker notes

Stepped, then manipulable (P6 tiers 2 and 3). The quadrant signs make the definition visible: covariance averages the sign of $(x - \bar{x})(y - \bar{y})$, so a cloud that is symmetric about the vertical line has the two signs in balance and averages to zero — however tightly the points sit on a curve.

Collect the prediction before switching. Almost everyone says "large": the dependence is total. The reading afterwards is ≈ 0 . That is the counterexample from the warning above, on screen, and it is the reason "uncorrelated" and "independent" must stay different words.



Takeaway

The joint distribution holds everything **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

Guiding Question: *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population

The entire group — all humans on earth, all transactions, all documents.

Its properties (μ, σ^2) are the truth we cannot observe.

Sample

The observations we actually have.

Its properties $(\bar{x}, s^2)$ are what we compute — estimates of the population's.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

Speaker notes

This slide is the hinge of the whole module: everything before it was deduction from a model, everything after is inference from data. The population/sample distinction is exactly the gap that statistics manages.



What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, IQR

Shape

3rd central moment
skewness, quantiles

Univariate statistics describe one random variable **multivariate** statistics describe $X_1, \dots, X_d$ together.

The moment ordering gives a clean mental index: where is it, how spread, how lopsided. Each has a robust and a non-robust representative — a distinction the next units make concrete.

Means

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Weighted mean

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i \right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Speaker notes

The sample mean estimates $E[X]$. Ask when the arithmetic mean is simply wrong: average speed over equal distances (harmonic), average growth rate (geometric). Choosing the mean is a modelling decision, not a formality.



Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$ -th value, for even n the average of the two middle values.

Robust against outliers — one extreme value moves the median by at most one position

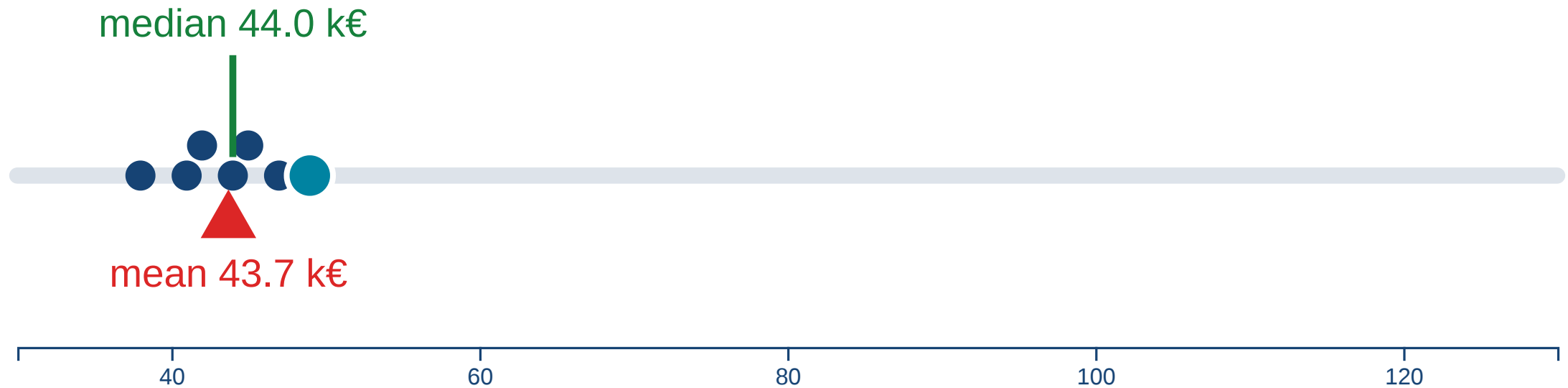
Costlier to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

The mean-versus-median contrast is the first robustness discussion of the course, and it returns in clustering (k-means vs. k-medoids) and in evaluation. Ask which one a salary report should quote, and why the answer is political.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?



Manipulable widget (P6 tier 3) with the prediction taken first, as the policy requires. The mean is drawn as the fulcrum the beam balances on, so the drag is visibly a lever: one value far out has to be balanced by everything else.

Drag it all the way and stop. The mean is now above six of the seven salaries — a "typical" value that almost nobody earns. The median has moved by one position. Then ask the political question: which number goes in the recruitment advert, and which one in the union's press release?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

The punchline: neither number is wrong, but only one is honest as a summary of "typical". This is the first appearance of the idea that a statistic encodes a question.

Mini-Exercise — Solution

Mean ≈ 73.9 , median = 44.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

The one-pass remark is the data-engineering link: streaming aggregations rely on it. Mention the numerical-stability caveat if the group is advanced.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\bar{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Dividing by $n - 1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Do not derive the correction give the intuition. A good probe: with $n = 1$, the sample variance is 0 (the point equals its own mean) — clearly not an estimate of population spread, and the corrected formula refuses to answer at all.

Takeaway

s^2 describes **the sample** $\hat{\sigma}^2$ estimates **the population**. The $n - 1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

The k -th q -**quantile** is the value x such that k/q of the values are smaller and $(q - k)/q$ are larger ($0 < k < q$).

Percentiles

$$q = 100$$

"the 12th percentile"

Quartiles

$$q = 4$$

$Q_1 = 25\text{th}$, $Q_2 = 50\text{th}$ (= median),

$Q_3 = 75\text{th}$ percentile

IQR

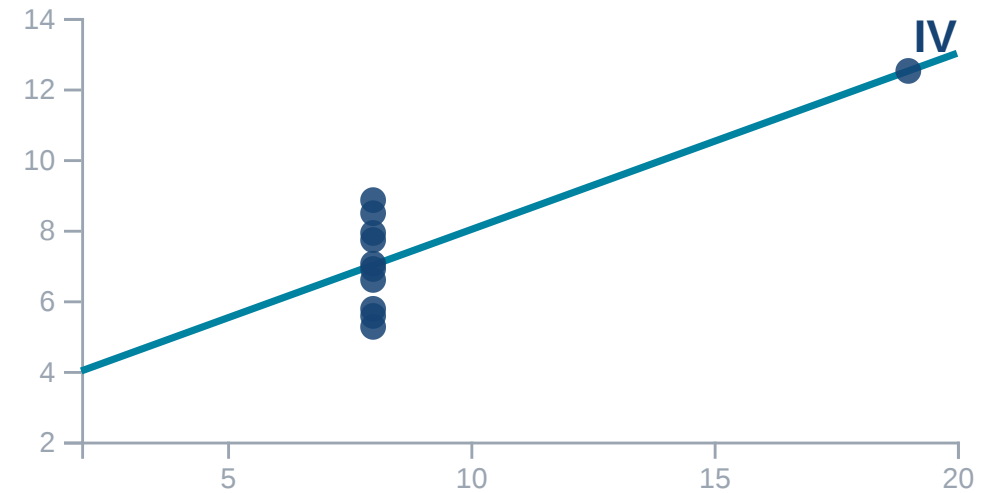
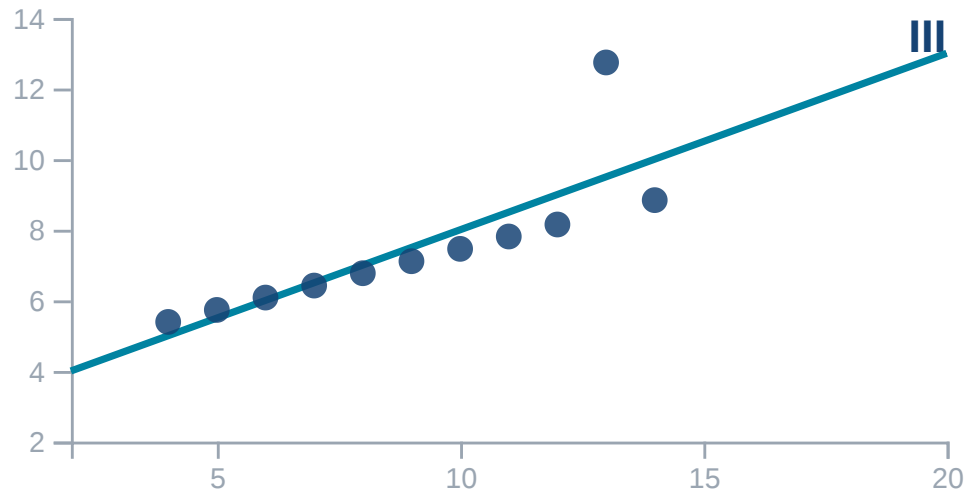
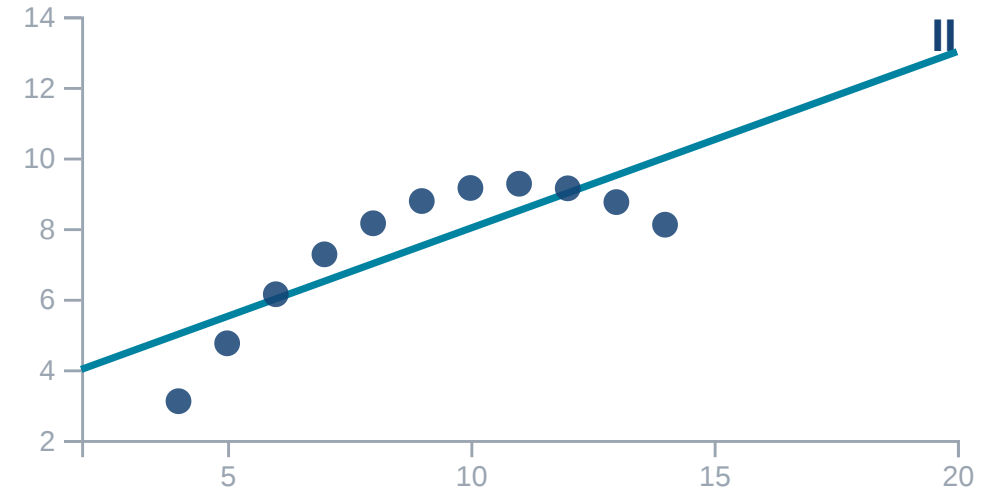
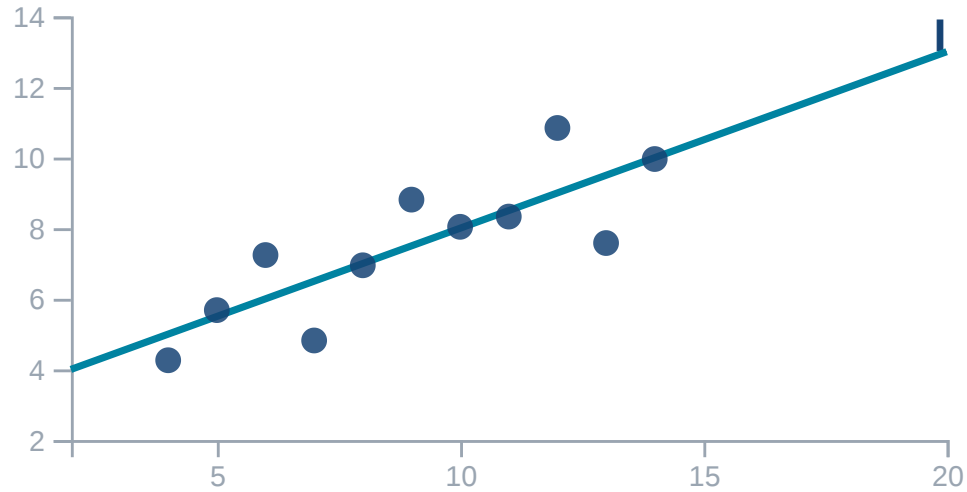
$$\text{IQR} = Q_3 - Q_1$$

the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Quantiles are the distribution-free alternative to moments: no assumption of symmetry, no sensitivity to tails. The boxplot is their standard graphical form — minimum, Q1, median, Q3, maximum, with outliers marked separately.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data



Stepped reveal (P6): one dataset per click, then the shared regression line. All four have the same mean, variance, correlation and least-squares line. Ask after the fourth click, before revealing: "these four have identical summary statistics — what does that tell you about summary statistics?" Anscombe (1973), the origin of "always plot your data".

What the Quartet Teaches

Summaries compress — and compression loses.

Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.

Visual inspection is not a soft alternative to statistics it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

This is the bridge to the data-exploration chapter, where visual analytics gets its own treatment. Stabilise the lesson as a working rule, not a slogan: every summary answers one question and is silent about the rest.

Module 7

Hypothesis Testing

Guiding Question: *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model
accuracy measured over n_A runs

Experiment B — new architecture
accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0 : the two means are equal — $\mu_A = \mu_B$

H_1 : they differ — $\mu_A \neq \mu_B$

Guiding question of the module, in the students' own working context. Every one of them has compared two model runs almost none has asked whether the difference survives resampling. Keep the example running through the unit.

The Idea

Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0 , reject H_0 .

We never prove H_1 . We only measure how badly H_0 fails to explain the data.

The asymmetry is the hardest idea in the unit. "Fail to reject" is not "accept". Ask what a court verdict of "not guilty" claims, and what it does not.



Test Statistic and p-Value

A **test statistic** $T = T(X_1, \dots, X_n)$ compresses the data so that its distribution **under** H_0 is known.

The **p-value** is the probability, under H_0 , of observing a statistic at least as extreme as the one we got.

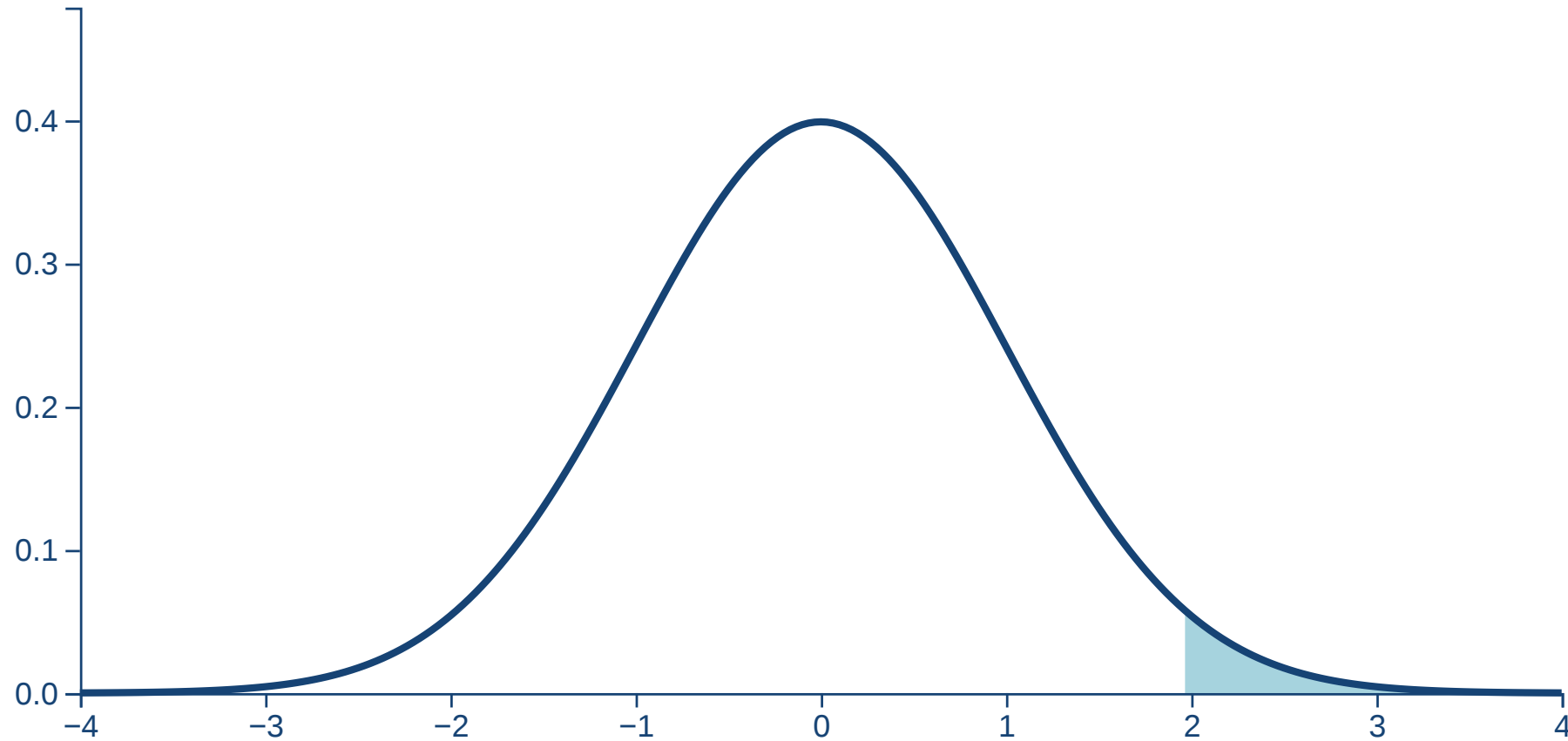
Small p-value $\rightarrow$ the data is surprising under H_0 .

The p-value is **not** the probability that H_0 is true. It is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.

The warning box is the Bayes unit returning: swapping the two conditionals is the same error as the disease test, now with a career's worth of published consequences. Make that link explicit — students remember it.

The Rejection Region

Normal $N(\mu = 0, \sigma = 1)$ — density



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

One click shades the tail. Emphasise: α is chosen *before* seeing the data, and it is the rate of false alarms we are willing to tolerate — a decision, not a discovery.



Two Ways to Be Wrong

Type I error — false alarm

Reject H_0 when it is true.

Probability α (we choose it).

Type II error — miss

Fail to reject H_0 when H_1 holds.

Probability β **power** = $1 - \beta$.

Lowering α raises β for a fixed sample size. More data is the only way to improve both.

Speaker notes

The trade-off is the practical content of the slide. In an ML context: a strict α means real improvements get dismissed; a lax α means noise gets published. Power analysis answers "how many runs do I need", which is the question people actually have.



Which Test?

One sample, known variance — z -test

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

One sample, unknown variance — t -test

$$T = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

Same standardised distance, estimated denominator → **heavier tails**.

Two samples, equal variances — pooled t

Use when variance homogeneity is credible (same pipeline, same noise).

Two samples, unequal variances — Welch's t

The default for ML A/B: different randomness induces different variances.

2x2 grid as a decision table. The single most useful takeaway for their own work is the bottom right: use Welch by default, because the equal-variance assumption is rarely justified across architectures.



Testing Variances

Single variance (normal model), χ^2 statistic with $n - 1$ degrees of freedom.

Is run-to-run variability within specification?

Two variances, $F = S_A^2 / S_B^2$.

If the variances differ, **do not pool** — go back to Welch's t for the means.

Variance tests matter in ML more than people think: a model that is equally accurate on average but wildly less stable across seeds is a different product.



Back to the A/B Question

Default

Welch's two-sample t on the mean difference

Care about stability

F -test on S_A^2 / S_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t) unequal variances change the test (Welch, not pooled).

Close by returning to the opening question with a concrete recommendation — the module started with a decision they have to make, and ends with the decision rule. Mention that the permutation test needs no distributional assumption at all, which is why it is a good fallback when in doubt.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.